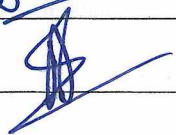


PART-A

Name of the Student: DINKESH PAL

Signature of the Student: 

Roll No.: 230123021

Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	4+2	9	20

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Let us write given system in FORM

$$Y = X\theta + \epsilon$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{3 \times 1} \quad X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}_{3 \times 2} \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}_{2 \times 1} \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}_{3 \times 1}$$

Design matrix

Objective function

$$O(\theta_1, \theta_2) = (Y - X\theta)^T (Y - X\theta)$$

So, $\hat{\theta}$ is LSE of θ matrix i.e. $\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$ where $\hat{\theta}_1$ & $\hat{\theta}_2$ is LSE of θ_1 & θ_2 respectively.

$$\Rightarrow \hat{\theta} = (X^T X)^{-1} X^T Y$$

$$\begin{aligned} X^T X &= \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}^T \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}_{2 \times 2} \end{aligned}$$

$$(X^T X)^{-1} = \frac{1}{27} \cdot \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} \frac{2}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{2}{9} \end{bmatrix}_{2 \times 2}$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} y_1 + y_2 + 2y_3 \\ y_1 - 2y_2 - y_3 \end{bmatrix}_{2 \times 1}$$

$$\hat{\theta} = (X^T X)^{-1} X^T y$$

$$= \begin{bmatrix} \frac{2}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{2}{9} \end{bmatrix} \begin{bmatrix} y_1 \cancel{y_1} + \cancel{y_2} + 2y_3 \\ y_2 \cancel{y_1} - 2y_2 - 2y_3 \end{bmatrix}_{2 \times 1} \quad \text{where } y_1 = x_1 + x_2 + 2x_3$$

$$y_2 = x_1 - 2x_2 - x_3$$

$$= \cancel{\frac{2}{9} y_1} \begin{bmatrix} \frac{2}{9} y_1 + \frac{1}{9} y_2 \\ \frac{1}{9} y_1 + \frac{2}{9} y_2 \end{bmatrix}_{2 \times 1}$$

$$\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} \frac{2}{9} y_1 + \frac{1}{9} y_2 \\ \frac{1}{9} y_1 + \frac{2}{9} y_2 \end{bmatrix}_{2 \times 1}$$

$$\hat{\theta}_1 = \frac{2}{9} y_1 + \cancel{\frac{2}{9} y_2} + \frac{4}{9} y_3 + \frac{1}{9} y_1 - \cancel{\frac{2}{9} y_2} - \frac{y_3}{9}$$

$$\hat{\theta}_1 = \frac{y_1}{3} + \frac{y_3}{3}$$

$$\hat{\theta}_2 = \frac{y_1}{9} + \frac{y_2}{9} + \frac{2y_3}{9} + \frac{2y_1}{9} - \frac{4y_2}{9} - \frac{2y_3}{9}$$

$$\hat{\theta}_2 = \frac{y_1}{3} - \frac{y_2}{3}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

~~Ans~~ . Since U_1 & U_2 are independent $\text{Cov}(U_1, U_2) = 0$.

$$U_1, U_2 \text{ are uniform R.V.} \quad E[U_1] = \frac{1}{2} \quad \text{Var}(U_1) = E[U_1^2] - E[U_1]^2 \\ = \frac{1}{3} - \frac{1}{4} \\ = \frac{1}{12}$$

$$E[U_2] = \frac{1}{2} \quad \text{Var}(U_2) = \frac{1}{12}$$

$$\rho_{x_1, x_2} = \frac{\text{Cov}(x_1, x_2)}{\sqrt{\text{Var}(x_1)} \sqrt{\text{Var}(x_2)}}$$

• Given :

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{3 \times 1}$$

$$x_1 : E[x_1] = E[U_1] = \frac{1}{2} \quad \text{Var}(x_1) = \frac{1}{12}$$

$$x_2 : E[x_2] = E[U_2] = \frac{1}{2} \quad \text{Var}(x_2) = \frac{1}{12}$$

$$x_3 : E[x_3] = E[U_1 + U_2] = 1 \quad \text{Var}(x_3) = \frac{1}{6}$$

(a) ρ Correlation Matrix $\rho = [\rho_{ij}]_{3 \times 3}$

$$\rho_{ij} = \frac{\text{Cov}(x_i, x_j)}{\sqrt{\text{Var}(x_i)} \sqrt{\text{Var}(x_j)}}$$

$$\rho_{11} = 1$$

$$\rho_{22} = 1$$

$$\rho_{33} = 1$$

$$\rho_{21} = 0$$

&

$$\rho_{31} = \frac{1}{\sqrt{2}}$$

$$\rho_{32} = \frac{1}{\sqrt{2}}$$

So, correlation matrix $\rho =$

$$\begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

$$\rho_{11} = \frac{\text{Var}(X_1)}{\text{Var}(X_1)} = 1$$

$$\rho_{22} = \frac{\text{Var}(X_2)}{\text{Var}(X_2)} = 1 \quad \rho_{33} = \frac{\text{Var}(X_3)}{\text{Var}(X_3)} = 1$$

$$\rho_{21} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1) \cdot \text{Var}(X_2)}} = \frac{\text{Cov}(U_1, U_2)}{\sqrt{\text{Var}(U_1) \cdot \text{Var}(U_2)}} = 0 = \rho_{12}$$

$$\begin{aligned} \rho_{31} &= \frac{\text{Cov}(U_1 + U_2, U_1)}{\sqrt{\text{Var}(U_1 + U_2) \cdot \text{Var}(U_1)}} = \frac{\text{Cov}(U_1, U_1) + \text{Cov}(U_2, U_1)}{\sqrt{\text{Var}(X_3) \cdot \text{Var}(X_1)}} \\ &= \frac{\frac{1}{6} + \frac{1}{6}}{\sqrt{\frac{1}{6} \cdot \frac{1}{6}}} = \frac{\frac{2}{6}}{\frac{1}{6}} = \frac{1}{3} = \rho_{13} \end{aligned}$$

$$\begin{aligned} \rho_{32} &= \frac{\text{Cov}(U_1 + U_2, U_2)}{\sqrt{\text{Var}(X_3) \cdot \text{Var}(X_2)}} = \frac{\text{Cov}(U_1, U_2) + \text{Cov}(U_2, U_2)}{\sqrt{\frac{1}{6} \cdot \frac{1}{6}}} \\ &= \frac{0 + \frac{1}{6}}{\frac{1}{6}} = \frac{1}{6} = \rho_{23} \end{aligned}$$

(b) Let Y_1 be first Principal Component (PC)

We know $Y_1 = C_1 X'$ & $\text{Var}(Y_1) = \lambda_1$

where (C_1, λ_1) is the pair of eigenvector & eigenvalue of matrix correlation matrix corresponding to largest eigenvalue. Also X' is standardized matrix of X .



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3. Let the matrix of distance be

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

- (a) (6 points) Cluster the four items using each of the following procedures:
- Complete linkage hierarchical procedure.
 - Average linkage hierarchical procedure.
- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

(or) (i) Complete linkage hierarchical procedure

$$d_{(uv)w} = \max(d_{uw}, d_{vw}) \quad \text{where } d_{ij} = \text{distance between } i \text{ \& } j$$

$$\begin{array}{c} \text{Init} \\ \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array} \xrightarrow{\begin{matrix} +d_{(12)3} \\ -d_{13} \\ -d_{23} \end{matrix}} \begin{array}{c} \begin{matrix} 3 & 4 & (12) \end{matrix} \\ \begin{matrix} 3 \\ 4 \\ (12) \end{matrix} \begin{bmatrix} 0 & & \\ 3.5 & 0 & \\ 11 & 5 & 0 \end{bmatrix} \end{array}$$

$$d_{(12)3} = \max(d_{13}, d_{23}) = 11$$

$$d_{(12)4} = \max(d_{14}, d_{24}) = 5$$

$$d_{(34)(12)} = \max(d_{3(12)}, d_{4(12)}) = 11$$

$$\begin{matrix} -d_{34} \\ -d_{43} \end{matrix} \downarrow + d_{(34)1}$$

$$\begin{array}{c} \begin{matrix} (12) & (34) \end{matrix} \\ \begin{matrix} (12) \\ (34) \end{matrix} \begin{bmatrix} 0 & \\ 11 & 0 \end{bmatrix} \end{array}$$

We repeat the process $N-1$ i.e. 3 times.

NOTE: Circled values in matrix are smallest values of d_{ij} in each iteration excluding diagonal entries.

(ii) Avg. linkage hierarchical procedure

$$d(uv)w = \frac{\sum_{i \in u} \sum_{j \in v} d_{ij}}{N_{uv} N_w}$$

$i \in uv$ cluster
 $j \in w$ cluster

$N_{uv} = \#(uv)$
 $N_w = \#(w)$
i.e. no. of items

Init

	1	2	3	4	
1	0				
2	①	0			
3	11	2	0		
4	5	3	3.5	0	

$\xrightarrow{\substack{+d_{12} \\ -d_1 \\ -d_2}}$

			3	4	(12)
3			0		
4			③.5	0	
(12)			6.5	4	0

$$d_{(12)3} = \frac{\sum_i \sum_j d_{ij}}{N_{12} N_3} = \frac{11 + 2}{2 \times 1} = \frac{13}{2} = 6.5$$

$$d_{(12)4} = \frac{\sum_i \sum_j d_{ij}}{N_{12} N_4} = \frac{5 + 3}{2 \times 1} = \frac{8}{2} = 4$$

$+d_{34}$

$\xrightarrow{\substack{+d_{34} \\ -d_3 \\ -d_4}}$

	(12)	(34)
(12)	0	
(34)	⑤.25	0

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$$d_{(12)(34)} = \frac{\sum_i \sum_j d_{ij}}{N_{12} N_{34}} = \frac{11 + 5 + 2 + 3}{2 \times 2} = \frac{21}{4} = 5.25$$

NOTE: Same we did this procedure $n-1$ i.e. 3 times
Circled values have some meaning here as well i.e.
Smallest at each iteration including diag. values.

CONTINUE

Q2 (b) ~~Q2~~ To get λ .

$$|S - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)((1-\lambda)(1-\lambda) - \frac{1}{2}) + \frac{1}{\sqrt{2}}(-\frac{1}{\sqrt{2}}(1-\lambda)) = 0$$

$$(1-\lambda)\left((1-\lambda)(1-\lambda) - \frac{1}{2} - \frac{1}{2}\right) = 0$$

$$(1-\lambda)(\lambda^2 - 2\lambda - \lambda) = 0$$

$$(1-\lambda)(\lambda^2 - 2\lambda) = 0$$

$$(1-\lambda)(\lambda)(\lambda - 2) = 0$$

So, 3 eigen value of S matrix are 2, 1, 0

$\lambda_1 = 2$ largest eigen value for first Principal Component.

~~Let $C_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1}$~~

~~To find $C_1 \Rightarrow (S - \lambda_1 I)C_1 = 0$~~

~~$$\begin{bmatrix} -1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$~~

$$\Rightarrow \textcircled{1} \quad -x + \frac{3}{\sqrt{2}} = 0$$

 $\Rightarrow$

$$x = \frac{3}{\sqrt{2}}$$

 $\Rightarrow \textcircled{2}$

$$-y + \frac{3}{\sqrt{2}} = 0$$

 $\Rightarrow$

$$y = \frac{3}{\sqrt{2}}$$

 $\Rightarrow \textcircled{3}$

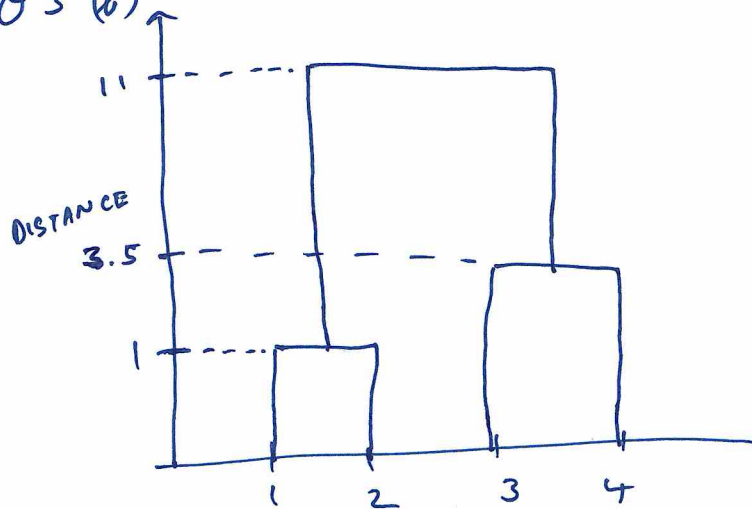
$$\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} - 3 = 0 \Rightarrow$$

$$\Rightarrow c_1' = \begin{bmatrix} \frac{3}{\sqrt{2}} & \frac{3}{\sqrt{2}} & 3 \end{bmatrix}$$

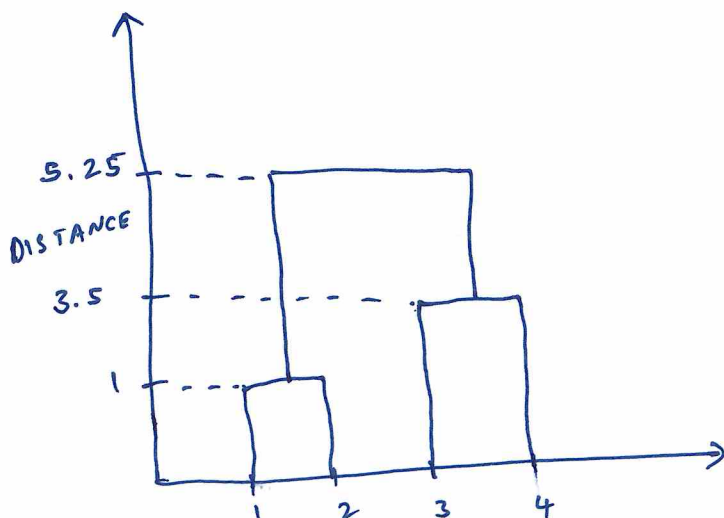
$$c_1' = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}^T$$

So, the variance of the first PC of based on correlation matrix is 2 with corresponding eigen vector $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}^T$

Q 3 (b)



Dendrogram for complete linkage hierarchical clustering procedure



Dendrogram for average linkage hierarchical clustering procedure

- We observe that in Average Linkage Hierarchical procedure, Smallest distance b/w 2 ^{different} clusters at each iteration was less than or equal to smallest distance b/w 2 diff. clusters in Complete linkage Hierarchical procedure for some iteration.
- Dendrogram height of full cluster converging into one big cluster is less for Average Linkage Hierarchical proc. ~~than~~ i.e. $5.25 < 11$.

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